An AI-driven Requirements Engineering Framework Tailored for Evaluating AI-Based Software

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***Abstract*—Requirements Engineering (RE) has been extensively refined for traditional software systems, but AI-based software (AIS)[1] introduces unique challenges that necessitate novel approaches.**

This paper addresses the gap in RE practices for AIS by proposing a framework that leverages partial specifications of domain concepts from RE and employs eXplainable AI (XAI) to verify AIS's perception of these specifications. The purpose of this framework is to demonstrate that systematically engineering AIS, according to RE practices, rather than fully relying on AI capabilities, will enhance the perception capabilities of resultant AIS.

This work aims to enhance RE4AI by offering a structured approach for managing and evaluating requirements specifications in AIS, ultimately leading to improved performance in these systems. Evaluation results showed that our framework improves AIS perception of variants of two domain concepts—pedestrian and aircraft—within the automotive and aviation domains.

***Index Terms*—RE4AI, Evaluating AI Software, RE and XAI**

I. Introduction

In engineering conventional software systems, requirements engineering (RE) practices have been extensively studied and refined. However, the development of AI-based software (AIS) introduces a host of unique and significant challenges to the RE process. While there exists a considerable body of literature on utilizing AI for managing RE activities (AI4RE), research specifically tailored to requirements engineering for AI (RE4AI) remains relatively scarce [10].

Establishing requirements is important as the requirements serve as benchmarks against which the AIS behaviors can be evaluated. Without them, it is challenging to assess whether the AI system meets its intended objectives effectively. In essence, the absence of clearly defined requirements and appropriate RE techniques undermines the reliability, performance, and overall success of AI systems in various applications. Therefore, establishing robust RE practices tailored for AI is essential to mitigate these risks and ensure the responsible development and deployment of AI technologies.

RE for AI presents unique challenges, necessitating a paradigm shift from conventional RE [25]. Unlike traditional RE, which focuses primarily on specifying requirements and ensuring they are met through a series of static verification and validation steps, RE4AI must go beyond this confines [26]. It requires a dynamic, iterative approach that continuously assesses and adapts specifications based on the AI model's evolving behaviors and learning outcomes. Given that AI-driven systems operate in probabilistic environments and often derive their behavior from complex data patterns rather than deterministic logic, traditional verification techniques fall short. In this context, RE4AI must address not only the specification of requirements but also how to verify and validate these specifications against an AIS perception and interpretation of the domain.

While there are existing works that have utilized traditional requirements engineering methods to generate a set of partial specifications for AI systems [5, 7, 9, 27, 38], this study advances these approaches by incorporating eXplainable AI (XAI) to systematically map and align an AI system's perception of these specifications with the initially established ones. The goal is to identify and address any deficiencies in either the specifications or the AI system's understanding of them.

This research introduces a framework that utilizes RE specifications to evaluate the extent to which a trained AI system has assimilated domain knowledge, while also explaining and enhancing the system's perceptual capabilities. We refer to our framework as W-AIS, which stands for (W)hitebox-like (transparent) improvement of AIS perception of domain concepts. This name reflects our RE-guided approach to enhancing AIS's perception of domain concepts while dynamically illustrating the evolution of the model's focus.

We demonstrated the effectiveness of our framework by showcasing improvements in AIS perception of domain concept variations across four key areas: 1) initially established specifications of the concept variants, 2) original specifications tied to the concept variants, 3) human-generated specifications, and 4) accurate detection of previously unseen instances of the concept variants.

To delineate the **scope** of our work, we concentrate specifically on RE for AIS which relies on vision-based perception of their operating environment, where decisions are derived from pixel-level data, such as images and video frames. Our focus is primarily on AI applications within the automotive and aviation domains, specifically addressing two concepts: *pedestrian* and *aircraft*. In this scope, we seek to answer the research questions below:

- RQ_1 : How can existing RE practices be adapted and effec-

[1]In this work, AI-based software (AIS) refers to software that relies exclusively on vision-based perception, meaning its understanding of the environment is derived solely from camera input.

tively integrated into the evaluation of AI-based software systems?

- ***RQ*₂ :** Does this integration improve the resultant AI-based software systems?

The contribution of this paper, in particular, is four folds:

1) **Integration of RE Practices into AI Development:** The framework offers a systematic approach to adapt RE methodologies to the unique challenges of AI, providing a structured framework for managing requirements in this context.
2) **Synergy of RE Practices with XAI Techniques:** The traditional RE practices are insufficient for AIS engineering. By integrating techniques from XAI, the framework facilitates the systematic identification of weaknesses in domain specifications and benchmarks.
3) **RE-empowered Enhancement of AI Perception:** The framework enables iterative targeted fine-tuning of AI perception capabilities. Prioritizing concept variants that are inadequately learned by the AI system, helps in mitigating overfitting and ensuring that the system can effectively recognize diverse instances within its domain.
4) **Image-Caption Datasest:** We systematically built two domain-specific image-text datasets for domain concepts of *pedestrian* and *aircraft*, containing 8,997 and 6,013 images, respectively, with corresponding descriptive captions.

Both datasets, the W-AIS framework, and the results of this research are made public in our shared repository[2].

II. Background

This section offers background information on key RE concepts relevant to our study and explores the challenges of applying RE principles to AIS, with a particular focus on those issues that directly pertain to our research.

A. Knowledge-based Requirements Engineering

Conceptualization refers to creating an abstract and simplified version of the world to be represented [18]. Every knowledge-based system is committed to some level of conceptualization [20], where concepts are defined abstractly using sub-class or specialization relationships. For example, in a bank application, terms like "customer" and "client" may refer to the same concept and need a formal definition. To elicit domain-specific requirements, software engineers first acquire domain knowledge to understand the domain elements (concepts), often terms found in user scenarios [23]. They then extract and classify properties of these concepts [4], structuring the gathered knowledge into a central domain specification knowledge base for future reuse [46]. For instance, in developing a pacemaker application in the medical domain, user manuals and other documents are parsed to extract necessary domain information about pacemaker requirements. With domain specifications available, requirements engineers create user scenarios and develop detailed requirements specifications, guiding the software design and implementation.

[2]https://github.com/AI-EnabledSoftwareEngineering-AISE/WAIS

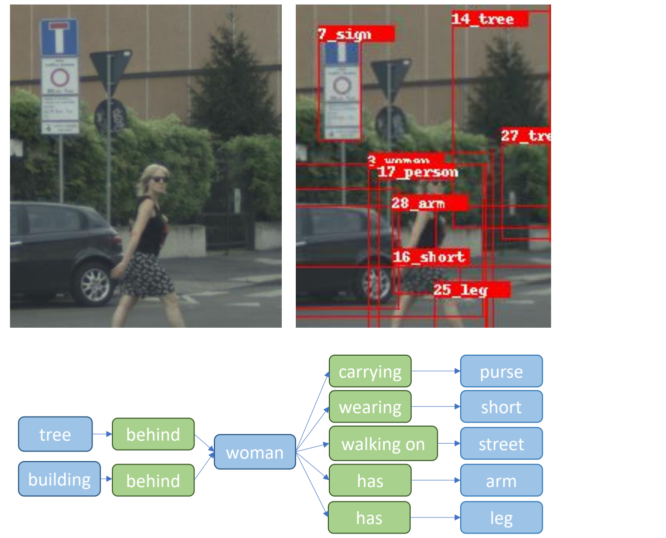


Fig. 1. Visual-based extraction of partial specifications for the domain concept "pedestrain"

To capture domain knowledge, there is a growing interest in using *ontologies*, which are "explicit specifications of a conceptualization" [22]. Ontologies are explicit due to the visibility of their classes and properties [18], and they can be logically reasoned and shared within a specific domain [24]. As standard representations of domain concepts and their relationships, ontologies facilitate automated reasoning. In RE, ontologies are employed to specify more complete and unambiguous requirements, perform consistency analysis, explicitly model domain knowledge, and manage requirements and changes [4, 29].

B. Domain Concepts in AIS

While the extraction of specifications for domain concepts is standard in RE, several challenges emerge when applying ontology-based RE to vision-based AIS:

1) The Intuitive Nature of AIS Domain Concepts: Unlike humans, who intuitively understand domain concepts through daily experiences and social interactions, AIS requires explicit, formal descriptions of these concepts. This creates a challenge for requirements engineers, who must provide comprehensive yet flexible specifications of these intuitive concepts. These descriptions must be detailed enough to capture the full range of existing concept variants, without being so rigid that they cannot generalize to future, unseen variants [42].

For example, in the automotive domain, the concept of a "pedestrian" is notoriously difficult to specify due to the vast range of unpredictable variations in pedestrian appearances, such as individuals using wheelchairs, scooters, or roller skates [28].

2) The Absence of a Reference Point: Without formal specifications of domain concepts—particularly those that are intuitive—it becomes difficult to assess AIS behavior. To what extent, for instance, should a pedestrian collision avoidance system be expected to recognize various pedestrian variants?

What about a pedestrian dressed in a Halloween hot-dog costume?

Evaluating a trained AIS for specific concepts typically starts by assessing the completeness of its training data. However, defining data sufficiency—especially for complex concepts like "pedestrian"—requires clear criteria. Key characteristics such as data correctness, completeness, and quantity need to be measured against a benchmark. Without a standard reference point, these assessments often depend on subjective human intuition, which can vary significantly across individuals. This lack of a standardized reference point makes evaluating AIS highly abstract.

3) More Data Is Not Necessarily Better: The size of a dataset is not a direct indicator of its quality. In fact, in data-driven modeling, overfitting is a common issue that can result from using excessively large datasets [53]. Overfitting occurs when a model becomes too tailored to the training data, leading to a decline in its ability to generalize to new, real-world scenarios. This deteriorates the model's trustworthiness and its performance when deployed. Therefore, simply increasing the quantity of data does not guarantee better results; careful consideration must be given to the quality, relevance, and diversity of the dataset to avoid such pitfalls.

TABLE I
TEXTUAL-BASED EXTRACTION OF PARTIAL SPECIFICATIONS FOR THE DOMAIN CONCEPT "PEDESTRAIN" REPRESENTED IN A TUPLE FORMAT OF *Subject-Predicate-Object*.

Src	Spec.	Sbj.	Predicate	Object
Textual	S_{1-3}	Pedestrian	has-a	{bike, wheelchair, safety...}
	S_{4-7}		does	{jaywalking, walking, crossing...}
	S_{8-10}		is-a	{handicapped, blind, drunk...}
Visual	S_{10-20}	Pedestrian	is	{people, person, woman, child...}
	S_{21-26}		wearing	{pant, shirt, shoe, jean, shorts...}
	S_{27-32}		carrying	{bag, umbrella, surfboard, paper...}
	S_{33-41}		has	{head, leg, arm, facial hair...}
	S_{42-46}		walking	{sidewalk, street, track, snow...}
	S_{47-50}		riding	{horse, bike, skateboard...}

III. DERIVING PARTIAL SPECIFICATIONS FOR AIS (RE-AIS)

Our framework aims to leverage existing domain knowledge to clearly and precisely define concepts before training AI systems, enabling the validation of these specifications within the AIS context. This approach is preferable to relying on randomly collected training data, which may be incomplete or unreliable. To establish these specifications, our framework adopts an ontology-based approach, which is particularly well-suited for the AIS engineering process for two primary reasons.

First, due to the intuitive nature of domain concepts needed for AIS's vision-based perception (*e.g.,* "pedestrian" in the automotive domain or "aircraft" in aviation), clear specifications are necessary to eliminate ambiguity and bias. Explicit delineation helps in understanding complex scenarios, such as differentiating a pedestrian riding a bike versus standing next to it. Second, given the wide range of variations within most AIS domain concepts, the specifications must capture these variants to ensure a comprehensive assessment of the AIS's perception capabilities.

TABLE II
HIGH-LEVEL TOPICS (T) OF *pedestrian* AND *aircraft* DOMAIN CONCEPTS.

T	Pedestrian Topics	Aircraft Topics
a	Physically-challenged Pedestrians	Unmanned Vehicles
b	Children Pedestrians	Historic Aircraft
c	Careless Pedestrians	General Aviation
d	Traffic Violations	Aircraft Types
e	Highways	Military Aviation
f	Protests	Commercial Planes
g	Culture	Aerospace Engineering
h	Detection Technologies	Airline Operations
i	Mobility Issues	Flight Incidents
j	Safety Issues	Pilot Experience
k	Infants	-
l	Vehicle Faults	-
m	Urban Areas	-
n	Pedestrian Behavior	-
o	Outdoor Activities	-

To establish the initial specifications, we utilized the REAIS framework described in [5], which semantically defines specifications for domain concepts by constructing extensive sets of ontologies derived from human knowledge sources. W-AIS builds upon the output of the REAIS framework. Consequently, this section provides a brief overview of this component, which automatically generates a set of domain-specific benchmarks (i.e., ontologies). For further details on REAIS, please refer to [5, 9].

The final ontology benchmarks contain large sets of semantically-important specifications of domain concepts, for which humans inherently have an intuition. The reason for exploiting the ontologies of domain specifications is to identify and minimize the gap between the perception of domain concepts, as humans experience and AIS inductively learn during an AIS training process.

To build the ontology benchmarks, REAIS searches through human knowledge sources, which contain information in Natural Language (NL) format, such as, generic and domain-specific encyclopedias, books, news feeds, social media repositories, dictionaries, legal documents, and articles, as well as those which consist of pixel-level information, such as image and video frame databases. REAIS adopts a wide range of Natural Language Processing (NLP) techniques [15, 49], to automatically mine textual sources of knowledge for *important* accompanying features of the concept.

Further, REAIS adopts various image processing techniques and convolutional neural networks (CNNs) [45, 52]. For instance, scene graph generation (SGG) techniques [45] translate pixel-level visual data, such as video and image information, to NL. The partial specifications of AIS targeted domain concepts are initially generated in the form of *(subject, predicate, object)* rules [6, 8, 9]. Figure 1 shows examples of partial specifications, retrieved from the given pedestrian instance.

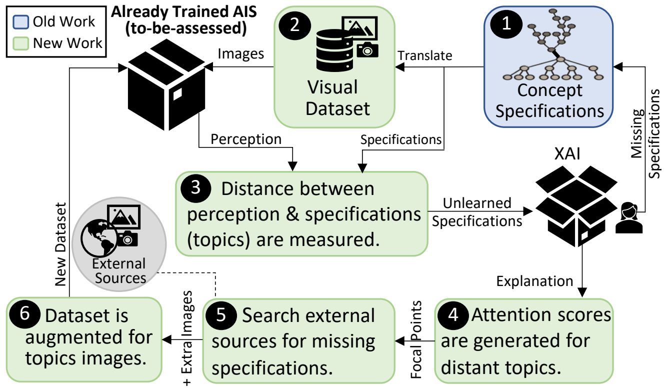


Fig. 2. W-AIS systematically identifies, targets, and explains the weaknesses of the AIS model's learning, as well as the specifications, and iteratively augments them.

Once the rules are populated, the *subject* is replaced with the hard-to-specify domain concept (here pedestrian), and the predicate is the form of a verb, describing a common relation between the subject and object, as REAIS specified. The object can be in the form of a noun or an adjective, representing an entity or description, which REAIS identified to be often accompanying the subject. Table I shows a few examples of partial specifications, retrieved from the given pedestrian instance.

Later, to facilitate the management, reuse, and visualization of the specifications, REAIS transforms the specifications into a network of connected graphs, in which nodes represent a set of relevant specifications with a hierarchy of higher-level *topics*. This paper adopts the term *topic* as a higher-level specification for a collection of the concept variants, identified in the benchmark to be semantically related to each other (shown in Table II). For instance, a few of the pedestrian topics (variants) include *"physically-challenged pedestrians"*, *"children pedestrians"*, and *"careless pedestrians"* (i.e., those jaywalking or reading papers while walking).

The final output is therefore in the form of several interconnected semantic webs, using the W3C Web Ontology Language (OWL), containing a set of high-level topics and associated lower-level descriptions of the concept's variants.

IV. An Overview of our Framework

Figure 2 presents a high-level visual overview of our framework's main components, while Algorithm 1 provides a formalization of the framework:

Figure 2: ❶ and Algorithm *(lines 2-3)*: As earlier explained, at different levels of granularity, the ontology benchmarks contain a large set of semantically important specifications of AIS domain concepts. Once built, the benchmarks are referred to by W-AIS, as a **minimum** set of requirements for AIS vision-based perception, which in turn enables our framework to estimate the extent, to which the under-assessment AIS is expected to perceive the variants of the concept.

Figure 2: ❷ and Algorithm: *(line 5)*: Since AIS perception is only from pixel-level information, to assess, our framework requires converting the benchmark topics into a set of corresponding instances of visual data, so that the topic is perceivable by AIS. For this purpose the ontology, which is originally in natural language and is stored in a graph-based structure, is required to be translated into a visual dataset, fairly representing the concept variants.

Later the AIS, whose perception is being analyzed, will be initially treated as a black box, receiving the set of translated images of each concept variant and generating its perception of the concept variants. To keep the process as transparent as possible for the human the AIS perception requires to be generated in a human-understandable language.

Figure 2: ❸ and Algorithm: *(line 6-10)*: To capture the AIS perception of the concept variants we use multi-modal generative models that use both image and text as prompts to produce text output.

To quantify the model's perception of each variant, we measure the distance between the model-generated specification of the concept variant and the human-derived specification in the benchmark. If this distance exceeds a set threshold, the topic is considered weakly perceived by the model and is labeled as an *"under-perceived specification"* by the AIS. The threshold value is initially determined through trial and error, with more sensitive safety domains requiring a higher threshold. When specifications are found to be missing from the benchmark, they are reviewed and added by a human to the initially established ontology to refine and enhance its completeness and accuracy.

Figure 2: ❹ and Algorithm: *(line 10-12)*: For the concept variants, which are not fully perceived by the model, the process selects an image, with the highest similarity distance, from the topic-related test-set images and displays the top features with the highest attention scores. This is visualized by a graphical representation, heat-maps [40, 44], as well as a combination of visual and textual explanations, as demonstrated in Figure 3.

By leveraging a state-of-the-art XAI approach, our framework intends to explain the AIS partial perception of the topic and assess the focal points of AIS perception against a more refined set of specifications of the variant in the established benchmarks. This explanation equips a human-in-the-loop with the necessary information to determine whether a large distance represents a missing specification, warranting its addition to the ontology or indicates an under-perceived topic, necessitating retraining of the model. The application of the XAI approach, in our framework, not solely allows for comprehending *why* an AIS model misperceived a variant but further helps to explain *when* a model's perception is improved by the framework.

For instance, in this case, the model mistakenly generated "pedestrian riding a bike". A mismatch, between the highly-attentive features of the model (*e.g.,* bike) and the detailed specifications (*e.g.,* pedestrian in wheelchair) of the topic (*e.g.,* physically-challenged pedestrians), represents that the model primarily focused on unnecessary features, and hence, the training process requires adjustments.

Figure 2: ❺ and Algorithm: *(line 13-16)*: To Adjust AIS

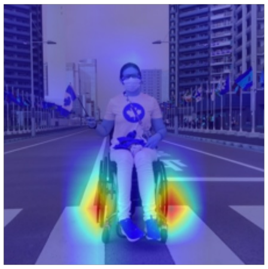


Fig. 3. The heatmap explanation for topic *"Physically-challenged Pedestrians"* with significant divergence between model's image explanation *"pedestrian riding a bike"* and the ontology's specification *"pedestrian in wheelchair"*.

training, only for the under-perceived variants and without impacting the AIS correct perception of the rest of the variants, our framework iteratively seeks through external sources for additional training data containing the refined specifications in the ontology.

Initially, the AIS's high-attention features (*e.g.,* bike) are mapped to the primary features of the corresponding topic (*e.g.,* physically-challenged pedestrians) in the established semantic ontology. If a mismatch is identified between the important features in the ontology (*e.g.,* wheelchair) and the model's primary focus (*e.g.,* bike), a batch of additional images relevant to the under-perceived specification (*e.g.,* pedestrian in wheelchair) is retrieved from user-specified external image sources.

Figure 2: ❻ and Algorithm: *(line 15)*: Finally, the last component in our framework, which is responsible for the adjustment, augments the dataset with additional images of the topic, including the semantically important features, and re-trains the model with the augmented dataset. This process is repeated until the average dissimilarity of the model's captions and the ground truth is reduced to below the specified threshold. It is possible that an iteration temporarily increases the distance of the model's perception of the variant from the ground truth, in particular, once the retrieved images for the retraining process do not fully capture the concept variant. However, the final results of our experiments showed that in general, the process leads to a smaller distance between what a model perceived of images of a concept variant and what is semantically specified for the variant. The framework ultimately generates a visual report of the AIS's focus shift.

The entire process ensures that AIS perception of its primary domain concepts, which was inductively learned during training, does not significantly deviate from what is established in the benchmarks, inferred based on human perception. This process ultimately improves AIS's visual understanding of domain concepts by enriching its data-driven knowledge with the knowledge-driven technique of domain analysis. The merge of the two learning processes yields a new generation of visually perceptive AIS, whose artificial intelligence is augmented with humans' domain knowledge, which in turn improves the machine's generalizability, robustness, and reliability.

V. Evaluation

We selected the state-of-the-art AI model, OFA [50] to evaluate our framework. We demonstrated how W-AIS improved its vision-based perception of two key concepts: *pedestrian* and *aircraft*. We chose OFA due to its capability to generate natural language captions for given images, using these captions to assess the model's contextual understanding of visual inputs. Specifically, we evaluated OFA's perception by measuring how closely its generated captions aligned with what was expected. We assessed that W-AIS improved AIS's perception through using four distinct metrics:

Conceptual Alignment: by generating captions for concept instances more closely aligned with the specified concept variations in our knowledge base.

Specification Alignment: To ensure the observed improvement was not merely a result of alignment with biased specifications, we showed that our framework also increased similarity to the original captions of the images of concept instances.

Human-Likeness: Due to potential shortcomings in the original captions, we further validated W-AIS by demonstrating an increased similarity to human-generated captions.

Performance Generalization: Finally, to confirm that the enhanced similarity translated into real-world performance gains, we demonstrated a more accurate recognition of unseen instances by AIS.

A. Translating Dataset and Measuring Distance

(Figure 2: ❷ and ❸) As explained earlier, the first step of our framework is to translate the established partial specifications into a visual dataset that accurately represents the specifications. To achieve this, we experimented with two different approaches to obtain visual translations of specifications: *Benchmark-based Search* (App_1) and *Web-based Search* (App_2).

In the first approach, visual translations were derived from state-of-the-art benchmark datasets specific to each domain concept, ensuring high-quality and relevant images. However, these datasets are inherently limited to the variations they contain, potentially overlooking uncommon or emerging instances. In contrast, the second approach utilized the entire web to construct a comprehensive visual ontology. This broader sourcing method captured a wider range of variations, resulting in improved image quality and greater relevance to the domain concepts.

Benchmark-based Approach (**App_1**) In this approach we selected a benchmark dataset, called Wikipedia-based Image Text Dataset (WIT) [43], which consists of 11.5 million image-text samples in 108 languages, containing 37.6 million distinct entities. Each WIT image is accompanied by multiple descriptive textual bodies, providing information about the image and page, from which the image is retrieved. For evaluation purposes, we selected a subset of images from this dataset that were relevant to REAIS's existing specifications of concept variants.

Algorithm 1 W-AIS: Steps 2-3 are the Authors' Previous Work.

Require: AIS model to be evaluated.
1: **function** EVALUATE(Model)
2: $Array\ S_c \leftarrow Specify\,(concept\ variants)$ ▷ Concept Specifications
3: $Array\ S_v \leftarrow Topic\_Model\,(S_c)$ ▷ Higher-level Specification of the concept (Topics)
4: **for each** $(variant \in \mathcal{S}_v)$ **do**
5: $Array\ I_v \leftarrow Translate\,(variant)$ ▷ Visual dataset
6: **for each** $(image \in \mathcal{I}_v)$ **do**
7: $Array\ C_i \leftarrow (blackbox)\,Evaluate\,(AIS, image)$ ▷ Captions
8: $Array\ Dis_v \leftarrow Measure\,(S_v, C_i)$ ▷ Average distance scores
9: **end for**
10: **while** $(av\_score \neq \mathbb{R}_{0-1})$ **do** ▷ A user-specified threshold
11: $image \leftarrow SelectRandom(I_v)$
12: $Array\ HA_e \leftarrow (whitebox)\,Evaluate\,(image)$ ▷ High-attention elements (Figure 4 (a) and (b))
13: **while** $(e \in \mathcal{H}A_e \neq f \in \mathcal{S}_v)$ **do**
14: $Array\ I_f \leftarrow Search\,(f, e)$ ▷ Search through external sources
15: $Augment\_Retrian\,(I_f, AIS)$ ▷ Augment with concept variant specification (Figure 4 (c) and (d))
16: **end while**
17: **end while**
18: **end for**
19: $report \leftarrow$ Report $(S_c, I_v, C_i, Sim_c, HA_e)$;
20: ▷ W-AIS output (Figure 4 (a), (b), (c) and (d)) **return** $report$
21: **end function**

We used a two-step process to select images from the WIT based on the similarity between the image captions with the topic specifications and the concepts. In the first step, we eliminated all the images for each topic, whose similarity to the topic and concept fell outside of the upper percentile of the entire dataset similarity. As a result, a subset of the images left was related to our concepts. Secondly, Given the similarity scores of image-concept and image-topic of the test-set images, similar to the previous process, we excluded the images, in which the image-topic similarity score fell outside of the upper quartile and within the first and second quartile relatively for the concepts pedestrian and aircraft. After removing duplicates, low-quality, and unsupported format (by the model), the final number of the WIT subset images was 7,789 and 2,727 image-text pairs for pedestrians and aircraft.

Web-based Approach (App_2) Due to the limitations observed in the benchmark datasets, we decided to create our dataset to better demonstrate the efficiency of W-AIS without the interference of noise and inconsistencies present in the existing datasets. We initially developed a script, which repeatedly queried the Google search engine with natural language specification of each concept variant in the ontology. The process then loaded the landing page of each returned image to retrieve the image, its captions, and a body of relevant textual description as the image context. The context later served to verify the image's relevancy to the searched topic in order to minimize the noise in this phase.

The search queries were built according to the specifications of each concept variant, established in the semantic benchmarks. The images were retrieved from the Google Image search engine using web scraping applications. To retrieve images of high quality, a size limit was incorporated to exclude images with a length and width of fewer than 200 pixels. Further, dropping entries with caption lengths of less than two words, a total of 40,625 and 13,845 images were retrieved for the concepts of pedestrian and aircraft, respectively.

Moreover, to avoid cascading the error of this step, we optimized this stage by removing the images with lower similarity scores to the concept variant. For this we leveraged a bimodal (text and image) model [37], to measure the similarity of each image to the corresponding concept variant specification and to discard the images which were of lower relevance to the topic. A bimodal image-text neural network is designed to process and understand both visual (image) and textual (text) data simultaneously and extract meaningful relationships between images and their corresponding textual descriptions, enabling them to perform tasks that require a deep understanding of both modalities.

We then removed the images, whose captions' similarity score to the concept variant fell outside of the upper quartile of all test-set images based on the base model performance. Another inclusion criterion for images was the image similarity score to the concept itself to be within the first and second quartile respectively for pedestrian and aircraft concepts. The condition was weakened for the aircraft images, to the second quartile, since a smaller number of the images were initially retrieved for the aircraft in comparison to the pedestrian concept. This led to a total of 3,483 and 1,668 image-text pairs for pedestrians and aircraft respectively which remained in the study.

After creating the visual datasets for each approach, in App_1 and App_2, the relevant images for each topic were separately fed into OFA, which generated captions based on its perception of the images. The distance between OFA's generated specifications and the ontology-based specifications was then measured for each topic.

To measure this distance, we selected semantic similarity, a metric successfully adopted in various applications such as information retrieval and text summarization [19]. To measure semantic similarity, we initially evaluated and compared three state-of-the-art universal sentence encoders, namely SBERT [39], USE [13], and InferSent [17] on 1% of our dataset. While the same data was fed to the three models, their performance significantly differed. The models nearly generated similar scores, however, the execution time was significantly shorter for SBERT. We believe this occurred since SBERT executes both the query and corpus embedding processes on the same GPU. This research hence eventually adopted SBERT to identify missing or under-represented concept variants.

A relatively higher average similarity score between OFA's output and the original captions of a variant's instances indicated a more mature perception of the variant by OFA. Conversely, a larger semantic distance reveals either a gap in OFA's perception relative to the established benchmarks or a missing specification in the ontology. In other words, this score represents the distance between the model's pixel-level perception and the human knowledge-based intuition of a concept variant. The larger the distance, the further the model deviates from the visual perception of the domain concept. However, this distance could also indicate a partial specification that was originally missing from the established ontology, which will be further resolved.

B. Providing Explanations and Model Retraining

(Figure 2: ❸, ❹ and ❺) Given the snapshot of the current literature on multimodal explainability, we selected CLIP, which utilizes the attention mechanism, focusing on the important visual and textual regions of the image and caption [14].

To retrain, additional images of the topic were fed to the model for targeted re-training. The additional images were retrieved by following either of the processes explained in App_1 and App_2.

The number of images to be retrieved was defined in the form of a lower bound for the similarity of the images to the topic and a minimum number of images in each iteration. As such, the number of retrieved images for each under-perceived topic was different, depending on the extent to which the topic was popular in the external sources. For evaluation purposes, we set the batch size of each re-training iteration to be equal to four, representing the number of images, necessary to be retrieved, for each identified under-perceived topic. The inclusion criteria, in the batch for each image, was that the similarity score of the caption with the topic specification was more than or equal to the average similarity of the present instances in the dataset. This included 4,154 and 1,125 in the web-based approach and 1,051 and 586 in the benchmark-based approach respectively for the pedestrian and aircraft concepts. The model was then retrained with additional images focused on the under-perceived topics.

We evaluated the model's ability to recognize instances of the concept across four scenarios: before applying W-AIS, after using W-AIS when domain specifications were translated

(a) A woman in a mask riding a bike .

(b) A woman pushing a motorcycle in the rain .

(c) A woman in a wheelchair with a face mask is on a crosswalk .

(d) A homeless woman walking in the rain .

Fig. 4. An AIS's generated captions and attention (a) (b) **before** and (c) (d) **after** W-AIS.

as explained in App_1, in App_2, and using a control group of randomly selected images from the WIT dataset. For a fair comparison, the number of images in the control group matched the quantity used in App_1. The results of these comparisons are discussed in detail in the following section.

Figure 4 provides an example of the W-AIS-generated report, illustrating how specifications-guided retraining of the model leads to improvements in its perception through shifting the model's attention toward features that are more aligned with the specifications of concepts. In (a), the model focused heavily on the cart's wheel. However, through W-AIS systematic retraining, the attention of key neurons gradually shifted from the wheel to more relevant *walking* which serves as a key discriminative feature for classifying an individual within the pedestrian class.

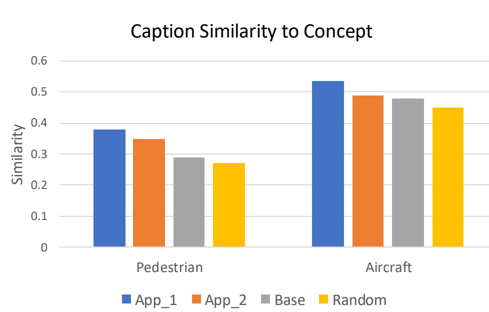


Fig. 5. The W-AIS improvement of semantic similarity between OFA-generated captions and the concepts.

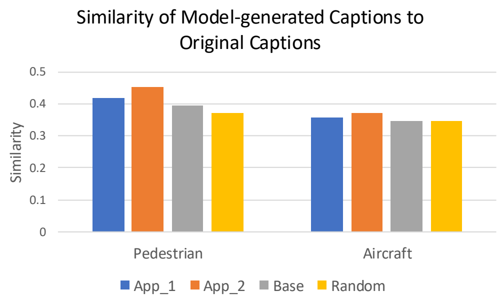


Fig. 6. The improvement of semantic similarity between OFA-generated and original captions.

VI. Results Discussion

We assessed our framework by demonstrating that the AIS's vision-based perception of variants of aircraft and pedestrian improved after utilizing our framework. This improvement was evidenced through four different metrics: (1) The increase in similarity between the generated captions and the refined specifications in the ontology. (2) The increase in similarity between the generated captions and the original captions of the images. (3) The increase in similarity between the generated captions and real human-generated captions. (4) The increase in correct classification of unseen instances. We measured and reported these metrics as an average across all concept variants to ensure that the retraining process not only improved the perception of under-perceived topics but also maintained the model's performance on initially well-perceived topics (relative to our threshold).

A. Improving Similarity to Ontology Specifications

We first measured how well the AIS-generated captions matched the refined specifications in our ontology. This similarity indicates how closely the AIS's understanding aligns with our established domain-specific knowledge. The high similarity suggests that the AIS effectively learns and incorporates domain-specific knowledge into its perception, making it more accurate and reliable in identifying and understanding domain concepts.

Figure 5 depicts semantic similarity between OFA-generated captions and variants of pedestrian concept on the left, and of aircraft concept on the right. The chart compares the performance of different models: $Base$, App_1, App_2, and $Random$, following the fine-tuning process.

For both concepts, the models fine-tuned on App_1 and App_2, consistently outperformed both the $Base$ and $Random$ models. For the pedestrian-related topics, the average similarity increased from 0.286 in $Base$ and 0.271 in $Random$ to 0.348 in App_2 and 0.378 in App_1. Similarly, for the aircraft-related topics, the average similarity increased from 0.476 in $Base$ and 0.450 in $Random$ to 0.489 in App_2 and 0.535 in App_1. These results demonstrate that fine-tuning the models on our curated dataset significantly enhances their performance compared to both the base model and the model trained on randomly collected images.

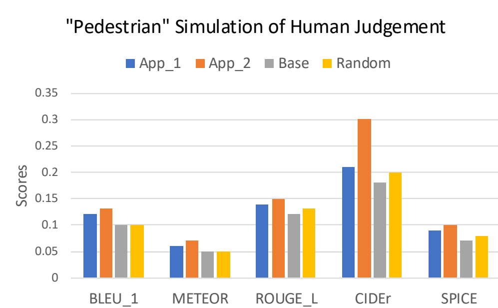
Please note that significance testing is not commonly employed in deep learning research due to the inherent high variance in model results, the availability of extensive datasets, and challenges in formulating clear hypotheses [11]. Furthermore, due to the Jeffreys-Lindley Paradox which is a situation in Bayesian statistics that highlights that a small p-value can contradict the posterior probability, leading to inconsistencies in interpreting results in comparative studies. Consequently, deep learning research tends to prioritize empirical evaluations over statistical significance testing [11].

B. Improving Similarity to Original captions

Next, we compared the AIS-generated captions to the original captions of the images. This metric shows how well the AIS can reproduce the descriptions initially associated with the images. By ensuring that the AIS-generated captions align closely with the original captions, we verify that the AIS retains the essential characteristics and context of the images, indicating a solid understanding of the visual content.

As shown in Figure 6, after the W-AIS applications in App_1 and App_2, the AIS-generated captions became more similar to the original image captions across all datasets. The results are presented in a bar chart, where the left bars indicate the semantic similarity for pedestrian-related images, and the right bars represent the similarity for aircraft-related images, comparing $Base$, App_1, App_2, and $Random$.

The similarity values for the pedestrian images increased from 0.395 in $Base$ and 0.369 in $Random$ to 0.454 in App_2 and 0.420 in App_1. Similarly, for the aircraft images, the similarity improved from 0.347 in $Base$ and 0.345 in $Random$ to 0.371 in App_2 and 0.358 in App_2. These results demonstrate that the W-AIS fine-tuning process significantly enhances the alignment between the AIS-generated captions and the original image captions compared to both the $Base$ model and the model trained on $Random$ collected data.



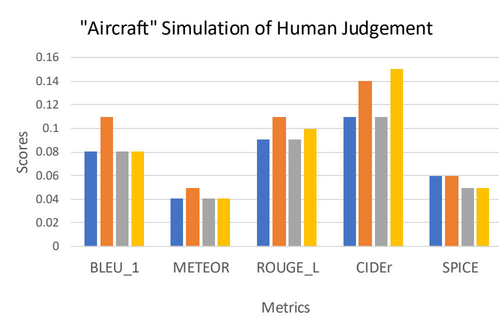


Fig. 7. The improvement of human-caption similarity for pedestrian and aircraft concepts using App_1 and App_2, compared to the $Base$ and $Random$ models.

C. Improving Similarity to Human-Generated Captions

We measured the similarity of AIS-generated captions to human-created ones, where higher similarity indicates that the AIS's interpretations are accurate and contextually appropriate, reflecting a more human-like perception. Since conducting user studies has the caveat of human bias, we initially measured this by using automated evaluation metrics. These automated metrics, including BLEU-4 [34], METEOR [1], ROUGE [30], CIDEr [47], and SPICE [3], have been shown to correlate highly with human responses and are widely adopted in literature due to their strong alignment with human evaluations.

As shown in Figure 7 for the pedestrian concept, App_2 achieved the best performance across all metrics. These results indicate that App_2 captions show higher similarity to human-generated captions compared to the $Base$ and other methods. App_1, while slightly behind App_2, still outperformed both the $Base$ and $Random$ datasets.

On average, for the pedestrian concept, App_2 showed approximately 67% improvement in CIDEr and 43% in SPICE over the $Base$, while App_1 showed around 17% and 29% improvement in CIDEr and SPICE, respectively. For the aircraft concept, the improvements were more modest overall compared with the Base, with App_2 showing around 27% improvement in CIDEr and 20% in SPICE, while App_1 improved only in SPICE by 20% and maintained the same performance as the Base in CIDEr.

The slightly smaller improvement observed in App_1 compared to App_2 is attributed to the quality of the original image captions used for fine-tuning. In App_2, the images were carefully curated and meticulously filtered by us, ensuring that the image-caption pairs were of high quality and captions closely aligned with the images, resulting in more accurate and contextually appropriate captions. Conversely, the images used in App_1 were sourced from the WIT dataset, which contained noisier and less focused captions, introducing inconsistencies that limited the fine-tuning performance. However, both App_1 and App_2 consistently outperformed the $Base$ and $Random$ approaches, demonstrating the effectiveness of the W-AIS framework in improving AIS perception of concept variants.

D. Improving Classification Accuracy

This evaluation specifically assesses the model's ability to detect key concept variants or topics as shown in Table II. Unlike prior evaluations, which focus on overall caption similarity, this evaluation focuses directly on how well the model identifies specific concept variants seen in an image. It thus provides a more targeted assessment of the model's interpretive capabilities, ensuring that it recognizes the essential variations of concepts in pedestrian and aircraft images.

For this evaluation, we enlisted three graduate students, uninvolved in our research, to annotate 620 pedestrian images and 517 aircraft images with the relevant concept variants (topics). In cases where the annotators disagreed on a tag, we implemented a voting system to determine its presence in the image. This approach ensured a fair consensus and enhanced the reliability of the annotations used as ground truth. Each pedestrian image received 4.4 tags on average, while each aircraft image was assigned 5 tags. These manual annotations served as the ground truth, ensuring a reliable reference for evaluating the concept variants.

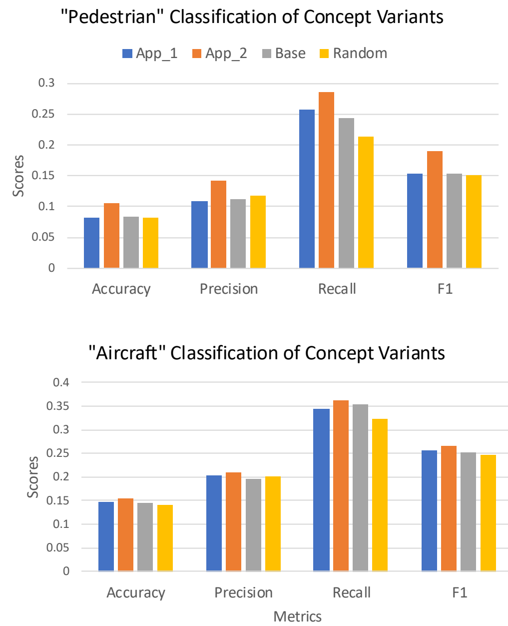


Fig. 8. The classification of concept variants evaluation results show the model's accuracy, precision, recall, and F1-score for detecting concept variants in pedestrian and aircraft images.

Subsequently, we deconstructed the captions generated by the model into individual terms and searched for semantically similar terms. We established a similarity threshold (in upper quartile) to determine the extent of correspondence between the model-generated terms and the terms in annotated tags. The prediction was classified as correct if the semantic similarity exceeded this predefined threshold. This method enabled us to identify true and false positives among the tags and calculate various performance metrics, including accuracy, precision, recall, and F_1 measures.

The advantage of this evaluation lies in its ability to assess the model's accuracy in detecting these concept variants. Evaluating the overlap between the concept variants in the ground truth and those referenced in the generated captions highlights the model's capacity for precise visual understanding.

As depicted in Figure 8, using a similarity threshold of 0.70, the results reveal significant differences in model performance for both concepts. For the pedestrian concept, the evaluation showed that App_2 performed best across all metrics, achieving an accuracy of 10.56%, precision of 14.32%, recall of 28.70%, and an F1-score of 19.11%. These numbers indicate that App_2 outperformed both the $Base$ and other methods.

Similarly, for the aircraft concept, App_2 once again led the performance metrics, with an accuracy of 15.35%, precision of 21.02%, recall of 36.25%, and an F1-score of 26.61%. These results underscore App_2's effectiveness at recognizing aircraft-related concept variants compared to other models like

App_1, which showed slightly lower performance in detecting these concepts. Once again, as earlier explained, the difference in performance between App_1 and App_2 is attributed to the nature of the datasets used for fine-tuning. App_2 was trained on a more curated set of image-text pairs, while App_1 used broader and more diverse image-text pairs as it was limited to the WIT dataset.

It is important to note that the generally low accuracy figures result from the model's emphasis on recognizing specific concept variants (e.g., "physically-challenged pedestrians" and "unmanned aerial vehicles") rather than merely identifying the broader categories (e.g., "pedestrian" and "aircraft"). The key takeaway is the improvement in these metric values, which reflect the model's enhanced capability to recognize these particular concept variants.

VII. Related Work

The field of SE for AI-based systems is rapidly evolving, but there is a significant gap in understanding the necessary SE practices for these systems. A systematic mapping study reveals that the field has emerged mostly since 2018, with dependability and safety being the most studied properties, while software maintenance remains underexplored. Data-related challenges throughout the software process indicate a need to adapt RE practices to AI development, especially in managing complex data requirements and ensuring system dependability and safety [32].

Effective RE for AI systems integrates traditional RE activities with AI-specific considerations, such as collaborative requirements analysis, which involves cross-domain collaborations and integrating domain knowledge to ensure AI systems can effectively respond to real-world scenarios [35]. Human-centered aspects, such as ethics, trust, fairness, and explainability, are also critical for ensuring AI systems are socially responsible and transparent [2].

Tailored RE practices are needed for machine learning (ML) systems due to their iterative and data-intensive nature. Frameworks like RE4HCAI focus on human-centered AI requirements, while structured approaches like GR4ML improve elicitation in complex domains like healthcare, which can be adapted for vision-based AI applications [21, 33].

Scenario-based requirements elicitation in XAI for fraud detection emphasizes understanding stakeholder needs for transparency and trust, crucial for vision-based AI systems [16]. Abstract architectures for explainable autonomy in hazardous environments highlight the importance of user trust by involving stakeholders in design and understanding their explanation needs [31].

Specifying requirements for AI systems involves detailed specifications that include evaluation measures and contextual information. Perspective-based approaches and structured meta-models help address complexities in vision-based AI systems [12, 51].

Integrating SE processes with AI development can enhance explainability and efficiency. Formal methods for reasoning and validation cover the entire ML lifecycle, ensuring AI systems meet their objectives. Validation approaches for AI-enabled software in sensitive domains stress the need for iterative processes and XAI to ensure compliance with intended use cases [36, 41].

Finally, practical tools like a catalog of concerns, validated by ML practitioners, ensure comprehensive coverage and enhance specification processes for vision-based AI systems [48].

VIII. Threats to Validity

Although we made an effort to avoid cascading errors throughout the process, by incorporating extra filtering stages, the partial pipeline architecture of the process may still include potential noise. The fact that we relied on CLIP and OFA to assess the similarity of text and image introduces the risk of adding the model's noise to the W-AIS framework. To minimize this threat, we selected the state-of-the-art models. A threat to the external validity is carrying out the experiments only in two domains (automotive and aviation). We designed a generalizable process, implemented a general framework, and referred to general knowledge sources.

While our framework incorporates a human-in-the-loop step that utilizes XAI to refine the initial specifications in the ontology, we opted to evaluate only the automated components with a fixed set of initial specifications, rather than the refined versions produced in each iteration. This allowed us to assess the foundational performance of the automated processes independently, providing a clearer understanding of their effectiveness and limitations. By focusing on the initial specifications, we can establish a solid baseline that will inform future enhancements and help identify where human input may be most valuable in subsequent iterations.

Finally, correct decisions by one ML model do not necessarily guarantee a correct decision at the system level, since the system as a whole, in the automotive application for instance, will also poll the other sensor inputs, such as radar and lidar input, to infer a decision. Yet, this work is the very first step toward system-level improvement.

IX. Conclusion

We introduced the W-AIS framework, which harnesses domain analysis and human semantic knowledge to iteratively enhance the visual perception of AI systems. Designed to address the challenges of RE4AI, the framework employs XAI to continuously refine established specifications. Evaluation results demonstrated that W-AIS has improved the AIS's vision-based perception of domain concepts, pedestrians and aircraft, within the automotive and aviation domains.

For future work, we plan to extend our framework by leveraging the association between the model's highly-attentive features and the domain specifications to directly shift the AIS's focus. This approach aims to eliminate the need for re-training, thereby reducing the computational costs typically involved in the refinement process.

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REFERENCES

[1] A. Agarwal and A. Lavie. Meteor: An automatic metric for mt evaluation with high levels of correlation with human judgments. *Proceedings of WMT-08*, 2007.

[2] K. Ahmad, M. Abdelrazek, C. Arora, M. Bano, and J. Grundy. Requirements engineering for artificial intelligence systems: A systematic mapping study. *Information and Software Technology*, page 107176, 2023.

[3] P. Anderson, B. Fernando, M. Johnson, and S. Gould. Spice: Semantic propositional image caption evaluation. In *European conference on computer vision*, pages 382–398. Springer, 2016.

[4] T. V. Avdeenko and N. V. Pustovalova. The ontology-based approach to support the requirements engineering process. In *International Scientific-Technical Conference on Actual Problems of Electronics Instrument Engineering (APEIE)*, volume 2, pages 513–518. IEEE, 2016.

[5] H. Barzamini and M. Rahimi. Cade: The missing benchmark in evaluating dataset requirements of ai-enabled software. In *2022 IEEE 30th International Requirements Engineering Conference (RE)*, pages 64–76. IEEE, 2022.

[6] H. Barzamini and M. Rahimi. CADE: The missing benchmark in evaluating dataset requirements of AI-enabled software. In *The 30th IEEE International Requirements Engineering Conference*. IEEE, 2022.

[7] H. Barzamini, M. Rahimi, M. Shahzad, and H. Alhoori. Improving generalizability of ml-enabled software through domain specification. In *2022 IEEE/ACM 1st International Conference on AI Engineering–Software Engineering for AI (CAIN)*, pages 181–192. IEEE, 2022.

[8] H. Barzamini, M. Rahimi, M. Shahzad, and H. Alhoori. Improving generalizability of ml-enabled software through domain specification. In *2022 IEEE/ACM 1st International Conference on AI Engineering–Software Engineering for AI (CAIN)*, pages 181–192. IEEE, 2022.

[9] H. Barzamini, M. Shahzad, H. Alhoori, and M. Rahimi. A multi-level semantic web for hard-to-specify domain concept, pedestrian, in ml-based software. *Requirements Engineering*, pages 1–22, 2022.

[10] H. Belani, M. Vukovic, and Ž. Car. Requirements engineering challenges in building ai-based complex systems. In *International Requirements Engineering Conference Workshops (REW)*, pages 252–255. IEEE, 2019.

[11] D. Berrar and W. Dubitzky. Should significance testing be abandoned in machine learning? *International Journal of Data Science and Analytics*, 7:247–257, 2019.

[12] D. M. Berry. Requirements engineering for artificial intelligence: What is a requirements specification for an artificial intelligence? In *International Working Conference on Requirements Engineering: Foundation for Software Quality*, pages 19–25. Springer, 2022.

[13] D. Cer, Y. Yang, S.-y. Kong, N. Hua, N. Limtiaco, R. S. John, N. Constant, M. Guajardo-Cespedes, S. Yuan, C. Tar, et al. Universal sentence encoder. *arXiv preprint arXiv:1803.11175*, 2018.

[14] H. Chefer, S. Gur, and L. Wolf. Generic attention-model explainability for interpreting bi-modal and encoder-decoder transformers. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, pages 397–406, 2021.

[15] G. G. Chowdhury. Natural language processing. *Annual review of information science and technology*, 37(1):51–89, 2003.

[16] D. Cirqueira, D. Nedbal, M. Helfert, and M. Bezbradica. Scenario-based requirements elicitation for user-centric explainable ai: A case in fraud detection. In *International cross-domain conference for machine learning and knowledge extraction*, pages 321–341. Springer, 2020.

[17] A. Conneau, D. Kiela, H. Schwenk, L. Barrault, and A. Bordes. Supervised learning of universal sentence representations from natural language inference data. *arXiv preprint arXiv:1705.02364*, 2017.

[18] D. Dermeval, J. Vilela, I. I. Bittencourt, J. Castro, S. Isotani, P. Brito, and A. Silva. Applications of ontologies in requirements engineering: a systematic review of the literature. *Requirements Engineering*, 21(4):405–437, 2016.

[19] W. S. El-Kassas, C. R. Salama, A. A. Rafea, and H. K. Mohamed. Automatic text summarization: A comprehensive survey. *Expert Systems with Applications*, 165:113679, 2021.

[20] R. Farooq. A conceptual model of knowledge sharing. *International Journal of Innovation Science*, 2018.

[21] A. Gjorgjevikj, K. Mishev, L. Antovski, and D. Trajanov. Requirements engineering in machine learning projects. *IEEE Access*, 2023.

[22] T. R. Gruber et al. A translation approach to portable ontology specifications. *Knowledge acquisition*, 5(2):199–221, 1993.

[23] J. Grundy, H. Khalajzadeh, and J. Mcintosh. Towards human-centric model-driven software engineering. In *ENASE*, pages 229–238, 2020.

[24] N. Guarino. *Formal ontology in information systems: Proceedings of the first international conference (FOIS'98), June 6-8, Trento, Italy*, volume 46. IOS press, 1998.

[25] U.-e. Habiba, M. Haug, J. Bogner, and S. Wagner. How mature is requirements engineering for ai-based systems? a systematic mapping study on practices, challenges, and future research directions. *arXiv preprint arXiv:2409.07192*, 2024.

[26] H.-M. Heyn, E. Knauss, A. P. Muhammad, O. Eriksson, J. Linder, P. Subbiah, S. K. Pradhan, and S. Tungal. Requirement engineering challenges for ai-intense systems development. In *2021 IEEE/ACM 1st Workshop on AI Engineering-Software Engineering for AI (WAIN)*, pages 89–96. IEEE, 2021.

[27] B. C. Hu, R. Salay, K. Czarnecki, M. Rahimi, G. Selim, and M. Chechik. Towards requirements specification for machine-learned perception based on human performance. In *2020 IEEE Seventh International Workshop*

on Artificial Intelligence for Requirements Engineering (AIRE), pages 48–51. IEEE, 2020.

[28] G. Li, Y. Yang, X. Qu, D. Cao, and K. Li. A deep learning based image enhancement approach for autonomous driving at night. *Knowledge-Based Systems*, 213:106617, 2021.

[29] R. Li, S. Ma, and W. Yao. Ontology-based requirements generation for credibility validation of safety-critical system. In *International Conference on Computer and Information Technology;*, pages 849–854. IEEE, 2015.

[30] C.-Y. Lin. ROUGE: A package for automatic evaluation of summaries. In *Text Summarization Branches Out*, pages 74–81, Barcelona, Spain, July 2004. Association for Computational Linguistics.

[31] M. Luckcuck, H. M. Taylor, and M. Farrell. An abstract architecture for explainable autonomy in hazardous environments. In *2022 IEEE 30th International Requirements Engineering Conference Workshops (REW)*, pages 108–113. IEEE, 2022.

[32] S. Martínez-Fernández, J. Bogner, X. Franch, M. Oriol, J. Siebert, A. Trendowicz, A. M. Vollmer, and S. Wagner. Software engineering for ai-based systems: a survey. *ACM Transactions on Software Engineering and Methodology (TOSEM)*, 31(2):1–59, 2022.

[33] S. Nalchigar, E. Yu, and K. Keshavjee. Modeling machine learning requirements from three perspectives: a case report from the healthcare domain. *Requirements Engineering*, 26:237–254, 2021.

[34] D. Nau, Y. Cao, A. Lotem, and H. Munoz-Avila. Shop: Simple hierarchical ordered planner. In *Proceedings of the 16th international joint conference on Artificial intelligence-Volume 2*, pages 968–973, 1999.

[35] Z. Pei, L. Liu, C. Wang, and J. Wang. Requirements engineering for machine learning: a review and reflection. In *International Requirements Engineering Conference Workshops (REW)*, pages 166–175. IEEE, 2022.

[36] A. Polzer, J. Fleiß, T. Ebner, P. Kainz, C. Koeth, and S. Thalmann. Validation of ai-based information systems for sensitive use cases: Using an xai approach in pharmaceutical engineering. In *HICSS*, pages 1–10, 2022.

[37] A. Radford, J. W. Kim, C. Hallacy, A. Ramesh, G. Goh, S. Agarwal, G. Sastry, A. Askell, P. Mishkin, J. Clark, et al. Learning transferable visual models from natural language supervision. In *International Conference on Machine Learning*, pages 8748–8763. PMLR, 2021.

[38] M. Rahimi, J. L. Guo, S. Kokaly, and M. Chechik. Toward requirements specification for machine-learned components. In *2019 IEEE 27th International Requirements Engineering Conference Workshops (REW)*, pages 241–244. IEEE, 2019.

[39] N. Reimers and I. Gurevych. Sentence-bert: Sentence embeddings using siamese bert-networks. *arXiv preprint arXiv:1908.10084*, 2019.

[40] M. T. Ribeiro, S. Singh, and C. Guestrin. " why should i trust you?" explaining the predictions of any classifier. In *Proceedings of the 22nd ACM SIGKDD international conference on knowledge discovery and data mining*, pages 1135–1144, 2016.

[41] F. Ritz, T. Phan, A. Sedlmeier, P. Altmann, J. Wieghardt, R. Schmid, H. Sauer, C. Klein, C. Linnhoff-Popien, and T. Gabor. Capturing dependencies within machine learning via a formal process model. In *International Symposium on Leveraging Applications of Formal Methods*, pages 249–265. Springer, 2022.

[42] R. Salay and K. Czarnecki. Using Machine Learning Safely in Automotive Software: An Assessment and Adaption of Software Process Requirements in ISO 26262. *ArXiv*, abs/1808.01614, 2018.

[43] K. Srinivasan, K. Raman, J. Chen, M. Bendersky, and M. Najork. Wit: Wikipedia-based image text dataset for multimodal multilingual machine learning. In *Proceedings of the 44th International ACM SIGIR Conference on Research and Development in Information Retrieval*, pages 2443–2449, 2021.

[44] H. Strobelt, S. Gehrmann, H. Pfister, and A. M. Rush. Lstmvis: A tool for visual analysis of hidden state dynamics in recurrent neural networks. *Transactions on visualization and computer graphics*, 24(1):667–676, 2017.

[45] K. Tang, Y. Niu, J. Huang, J. Shi, and H. Zhang. Unbiased scene graph generation from biased training. In *Conference on Computer Vision and Pattern Recognition*, pages 3716–3725. IEEE, 2020.

[46] S. Vasanthapriyan, J. Tian, and J. Xiang. A survey on knowledge management in software engineering. In *International Conference on Software Quality, Reliability and Security-Companion*, pages 237–244. IEEE, 2015.

[47] R. Vedantam, C. Lawrence Zitnick, and D. Parikh. Cider: Consensus-based image description evaluation. In *Proceedings of the IEEE conference on computer vision and pattern recognition*, pages 4566–4575, 2015.

[48] H. Villamizar, M. Kalinowski, et al. A catalogue of concerns for specifying machine learning-enabled systems. *arXiv preprint arXiv:2204.07662*, 2022.

[49] H. M. Wallach. Topic modeling: beyond bag-of-words. In *Proceedings of the 23rd international conference on Machine learning*, pages 977–984, 2006.

[50] P. Wang, A. Yang, R. Men, J. Lin, S. Bai, Z. Li, J. Ma, C. Zhou, J. Zhou, and H. Yang. Ofa: Unifying architectures, tasks, and modalities through a simple sequence-to-sequence learning framework. *CoRR*, 2022.

[51] X. Wang and M. Miao. A framework for requirements specification of machine-learning systems. In *Conference on Software Engineering and Knowledge Engineering (SEKE)*, volume 143, 2022.

[52] P. Xu, X. Chang, L. Guo, P.-Y. Huang, X. Chen, and A. G. Hauptmann. A survey of scene graph: Generation and application. *IEEE Trans. Neural Netw. Learn. Syst*, 2020.

[53] X. Ying. An overview of overfitting and its solutions. In *Journal of Physics: Conference Series*, volume 1168, page 022022. IOP Publishing, 2019.